

Diachronic Sample Integration: Robust Tail-Risk Estimation with Generative Models

Appendix

APPENDIX

This appendix provides the full assumptions, formal theorem, and proofs for the finite-budget theory introduced in the main paper. The proof uses a two-stage decomposition: first compare the checkpoint mixture $\bar{P}_S$ with the truth P^* , and then compare the empirical DSI estimator with the fixed mixture $\bar{P}_S$. This avoids assuming a direct empirical plug-in linearization around P^* .

a) *Mapping to main-paper numbering.*: Theorem 1 in this appendix is the full two-stage statement corresponding to Main Paper Theorem 1 (Finite-budget DSI MSE expansion). Corollary 1 corresponds to Main Paper Corollary 1 (Mechanistic regimes), and Proposition 1 corresponds to Main Paper Proposition 1 (Failure modes). Appendix-only results (for example, Lemma 1, Proposition 2, and Proposition 3) are supplementary technical statements used in the proofs.

A. Assumptions and Full Theorem Statement

Assumption 1 (Local tail regularity): F^* has a unique α -quantile q^* and is continuously differentiable on $I_\delta = [q^* - \delta, q^* + \delta]$, with density $f^*(q^*) > 0$ and $\inf_{x \in I_\delta} f^*(x) \geq c > 0$. When second-order remainders are invoked, f^* is locally Lipschitz on I_δ .

Assumption 2 (Post-burn-in checkpoint-error process): Let $X_t \sim P_t$ and $X^* \sim P^*$. Define $e_t(x) = F_t(x) - F^*(x)$,

$$U_t^q = e_t(q^*), \quad (1)$$

$$U_t^{ES} = \int_{-\infty}^{q^*} e_t(x) dx = \mathbb{E}[(q^* - X_t)_+] - \mathbb{E}[(q^* - X^*)_+]. \quad (2)$$

After burn-in, assume $U_t^q = b_q + \xi_t^q$ and $U_t^{ES} = b_{ES} + \xi_t^{ES}$, where b_q, b_{ES} are persistent bias terms and ξ_t^q, ξ_t^{ES} are mean-zero weakly stationary fluctuations with variances $\sigma_q^2, \sigma_{ES}^2$ and autocorrelations $\rho_q(h), \rho_{ES}(h)$.

Assumption 3 (Sampling and lower-tail moments): Assume $K_S \leq N$ and set $m = \lfloor N/K_S \rfloor$ and $N_S = K_S m$. Conditional on the selected checkpoint distributions $\{P_t : t \in S\}$, the generated samples $\{X_{t,i} : t \in S, 1 \leq i \leq m\}$ are independent with $X_{t,i} \sim P_t$. For all quantiles appearing below, the lower-tail variables $(q - X_t)_+$ have uniformly bounded second moments, and the analogous condition holds under P^* .

Assumption 4 (Local mixture regularity): Let

$$q_S := q_\alpha(\bar{P}_S), \quad (3)$$

$$\bar{e}_S(x) := \bar{F}_S(x) - F^*(x), \quad (4)$$

$$\eta_S := \sup_{x \in I_\delta} |\bar{e}_S(x)|. \quad (5)$$

Define the two leading checkpoint-mixture errors

$$\bar{U}_{q,S} := \bar{F}_S(q^*) - F^*(q^*), \quad (6)$$

$$\bar{U}_{ES,S} := \int_{-\infty}^{q^*} (\bar{F}_S(x) - F^*(x)) dx, \quad (7)$$

and the local oscillation of the mixture error around q^* ,

$$\omega_S(r) := \sup_{\substack{x \in I_\delta \\ |x - q^*| \leq r}} |\bar{e}_S(x) - \bar{e}_S(q^*)|. \quad (8)$$

Along the selected post-burn-in windows, the mixture remains in a local perturbation neighbourhood of P^* : $q_S \in I_\delta$, $\bar{F}_S(q_S) = \alpha$, and $\bar{F}_S$ has a density $\bar{f}_S$ in a neighbourhood of q_S with $\bar{f}_S(q_S) \geq c_0 > 0$. The local oscillation and second-order terms are negligible relative to the leading checkpoint-mixture errors:

$$\mathbb{E}[(\omega_S(\eta_S/c) + \eta_S^2)^2] + \mathbb{E}[|\bar{U}_{q,S}|(\omega_S(\eta_S/c) + \eta_S^2)] = o(\mathbb{E}[\bar{U}_{q,S}^2]), \quad (9)$$

and

$$\mathbb{E}[\eta_S^4] + \mathbb{E}[|\bar{U}_{ES,S}| \eta_S^2] = o(\mathbb{E}[\bar{U}_{ES,S}^2]). \quad (10)$$

Assumption 4 is a local perturbation condition: it requires the selected post-burn-in mixture to remain in a neighbourhood where VaR and ES admit stable first-order expansions.

Assumption 5 (Standard stratified plug-in linearization): Conditional on the selected checkpoint distributions, the empirical DSI estimators satisfy the usual Bahadur/plug-in linearizations around the fixed mixture $\bar{P}_S$:

$$\widehat{q}_{\alpha,S,N}^{DSI} - q_S = -\frac{\widehat{F}_{S,N}(q_S) - \alpha}{\bar{f}_S(q_S)} + r_{q,S,N}^{sam}, \quad (11)$$

$$\widehat{\text{ES}}_{\alpha,S,N}^{DSI} - \text{ES}_\alpha(\bar{P}_S) = -\frac{1}{\alpha} \int_{-\infty}^{q_S} (\widehat{F}_{S,N}(x) - \bar{F}_S(x)) dx + r_{ES,S,N}^{sam}, \quad (12)$$

with $\mathbb{E}[(r_{q,S,N}^{sam})^2 \mid \{P_t\}] = o(N_S^{-1})$ and $\mathbb{E}[(r_{ES,S,N}^{sam})^2 \mid \{P_t\}] = o(N_S^{-1})$ along the finite-budget sequences considered.

These conditions hold under standard local density and moment assumptions for independent, stratified triangular arrays with positive density at the target mixture quantile. They are stated explicitly to separate sampling regularity from trajectory regularity in Assumption 4.

a) *Full theorem statement.*: For an autocorrelation function ρ , define

$$\kappa_\rho(K_S, M) = 1 + 2 \sum_{\ell=1}^{K_S-1} \left(1 - \frac{\ell}{K_S}\right) \rho(\ell M), \quad (13)$$

$$K_{\text{eff},\rho}(K_S, M) = \frac{K_S}{\kappa_\rho(K_S, M)}, \quad (14)$$

assuming $\kappa_\rho(K_S, M) > 0$. Define the conditional sampling variances at the mixture quantile

$$A_{q,N}^{\text{mix}}(S) = \frac{1}{K_S^2 m} \sum_{t \in S} F_t(q_S)(1 - F_t(q_S)), \quad (15)$$

$$A_{ES,N}^{\text{mix}}(S) = \frac{1}{K_S^2 m} \sum_{t \in S} \text{Var}_{P_t}[(q_S - X_t)_+]. \quad (16)$$

For comparison, write $A_{q,N}(S)$ and $A_{ES,N}(S)$ for the same two quantities with q_S replaced by q^* . Unless explicitly conditioned, expectations below are taken over both the post-burn-in checkpoint-error process and the synthetic samples drawn conditionally from the selected checkpoints.

Theorem 1 (Main Paper Theorem 1 (full two-stage statement)): Suppose Assumptions 1-5 hold. Define

$$L_q^{\text{traj}} = b_q^2 + \frac{\sigma_q^2}{K_{\text{eff},q}(K_S, M)}, \quad L_{q,N}^{\text{sam}} = \mathbb{E} \left[\frac{A_{q,N}^{\text{mix}}(S)}{f_S(q_S)^2} \right], \quad (17)$$

$$L_{ES}^{\text{traj}} = b_{ES}^2 + \frac{\sigma_{ES}^2}{K_{\text{eff},ES}(K_S, M)}, \quad L_{ES,N}^{\text{sam}} = \mathbb{E} A_{ES,N}^{\text{mix}}(S). \quad (18)$$

Then

$$\mathbb{E} \left[\left(\widehat{q}_{\alpha,S,N}^{\text{DSI}} - q^* \right)^2 \right] = \frac{L_q^{\text{traj}}}{f^*(q^*)^2} + L_{q,N}^{\text{sam}} + \mathcal{R}_{q,N}, \quad (19)$$

$$\mathbb{E} \left[\left(\widehat{\text{ES}}_{\alpha,S,N}^{\text{DSI}} - \text{ES}_\alpha(P^*) \right)^2 \right] = \frac{(L_{ES}^{\text{traj}} + L_{ES,N}^{\text{sam}})}{\alpha^2} + \mathcal{R}_{ES,N}, \quad (20)$$

where

$$\mathcal{R}_{q,N} = o \left(\frac{L_q^{\text{traj}}}{f^*(q^*)^2} \right) + o(N_S^{-1}), \quad (21)$$

$$\mathcal{R}_{ES,N} = o \left(\frac{L_{ES}^{\text{traj}}}{\alpha^2} \right) + o(N_S^{-1}). \quad (22)$$

If, additionally, $\bar{f}_S(q_S) = f^*(q^*) + o_{\mathbb{P}}(1)$ and the stratified variances at q_S are first-order equivalent to those at q^* , this reduces to the simpler common-denominator form with $A_{q,N}(S)$ and $A_{ES,N}(S)$ evaluated at q^* . In the notation of Main Paper Theorem 1, $L_{q,N}^{\text{sam}} = L_{q,N}^{\text{sam}}$ and $L_{ES,N}^{\text{sam}} = L_{ES,N}^{\text{sam}}/\alpha^2$.

Corollary 1 (Main Paper Corollary 1 (Mechanistic regimes)): Theorem 1 identifies three regimes: (i) a *variance-reducible* regime where the corresponding bias term is small and K_{eff} grows, (ii) a *correlation-limited* regime where checkpoint fluctuations are present but highly autocorrelated, so $K_{\text{eff}} \ll K_S$, and (iii) a *bias-limited* regime where b^2 dominates and DSI cannot remove the leading error term.

Proposition 1 (Main Paper Proposition 1 (Failure modes)): DSI cannot correct perfectly correlated checkpoint errors, shared directional lower-tail misspecification across all selected checkpoints, or persistent bias floors. In particular, if $\rho(\ell M) = 1$ for all $\ell = 1, \dots, K_S - 1$, then $K_{\text{eff}} = 1$. If, for some sign $s \in \{-1, 1\}$ and nonnegative function a , $s\{F_t(x) - F^*(x)\} \geq a(x)$ for all $t \in S$ on a tail interval, then $s\{\bar{F}_S(x) - F^*(x)\} \geq a(x)$ on that interval. Even when $K_{\text{eff}} \rightarrow \infty$ and $N \rightarrow \infty$, the first-order VaR and ES MSE floors remain $b_q^2/f^*(q^*)^2$ and b_{ES}^2/α^2 , respectively.

B. Auxiliary Statements

Lemma 1 (Smooth first-order tail-functional expansions): Let G be a continuous CDF with finite lower-tail first moment, $q_G = q_\alpha(G) \in I_\delta$, and $G(q_G) = \alpha$. Write $g = G - F^*$, $\Delta = q_G - q^*$, $\eta = \sup_{x \in I_\delta} |g(x)|$, and

$$\omega_G(r) := \sup_{\substack{x \in I_\delta \\ |x - q^*| \leq r}} |g(x) - g(q^*)|. \quad (23)$$

Under Assumption 1, $|\Delta| \leq \eta/c$ and

$$q_\alpha(G) - q^* = -\frac{g(q^*)}{f^*(q^*)} + O(\omega_G(\eta/c) + \eta^2), \quad (24)$$

while

$$\text{ES}_\alpha(G) - \text{ES}_\alpha(P^*) = -\frac{1}{\alpha} \int_{-\infty}^{q^*} g(x) dx + O(\eta^2). \quad (25)$$

Proposition 2 (Checkpoint-error averaging): Under Assumption 2,

$$\mathbb{E} \left[\left(\frac{1}{K_S} \sum_{t \in S} U_t^q \right)^2 \right] = b_q^2 + \frac{\sigma_q^2}{K_{\text{eff},q}(K_S, M)}, \quad (26)$$

and

$$\mathbb{E} \left[\left(\frac{1}{K_S} \sum_{t \in S} U_t^{ES} \right)^2 \right] = b_{ES}^2 + \frac{\sigma_{ES}^2}{K_{\text{eff},ES}(K_S, M)}. \quad (27)$$

Proposition 3 (Stratified DSI sampling identities): Under Assumption 3, condition on the selected checkpoint distributions. For any fixed $x \in \mathbb{R}$,

$$\mathbb{E}[\widehat{F}_{S,N}(x) \mid \{P_t : t \in S\}] = \bar{F}_S(x), \quad (28)$$

and

$$\text{Var}[\widehat{F}_{S,N}(x) \mid \{P_t : t \in S\}] = \frac{1}{K_S^2 m} \sum_{t \in S} F_t(x)(1 - F_t(x)). \quad (29)$$

If $m = N/K_S$ exactly, then this variance is at most $\bar{F}_S(x)(1 - \bar{F}_S(x))/N$. Moreover, for any fixed threshold q with finite lower-tail second moments,

$$Z_{ES,N}(q) := \int_{-\infty}^q (\hat{F}_{S,N}(x) - \bar{F}_S(x)) dx \quad (30)$$

satisfies

$$\mathbb{E}[Z_{ES,N}(q) \mid \{P_t : t \in S\}] = 0, \quad (31)$$

and

$$\text{Var}[Z_{ES,N}(q) \mid \{P_t : t \in S\}] = \frac{1}{K_S^2 m} \sum_{t \in S} \text{Var}_{P_t}[(q - X_t)_+]. \quad (32)$$

C. Proof of Lemma 1

Because $G(q_G) = \alpha$ and $F^*(q^*) = \alpha$,

$$0 = G(q_G) - F^*(q^*) = F^*(q^* + \Delta) - F^*(q^*) + g(q_G). \quad (33)$$

The lower density bound in Assumption 1 implies

$$c|\Delta| \leq |F^*(q^* + \Delta) - F^*(q^*)| = |g(q_G)| \leq \eta, \quad (34)$$

so $|\Delta| \leq \eta/c$. A Taylor expansion of F^* around q^* gives

$$0 = f^*(q^*)\Delta + g(q^*) + \{g(q_G) - g(q^*)\} + O(\Delta^2). \quad (35)$$

Since $|\Delta| \leq \eta/c$, the oscillation term is bounded by $\omega_G(\eta/c)$ and $\Delta^2 = O(\eta^2)$. Hence

$$\Delta = -\frac{g(q^*)}{f^*(q^*)} + O(\omega_G(\eta/c) + \eta^2), \quad (36)$$

which proves the VaR expansion.

For ES, use the quantile-integral/lower-tail CDF identity, valid for continuous distributions with finite lower-tail first moment,

$$\text{ES}_\alpha(G) = q_G - \frac{1}{\alpha} \int_{-\infty}^{q_G} G(x) dx, \quad (37)$$

$$\text{ES}_\alpha(P^*) = q^* - \frac{1}{\alpha} \int_{-\infty}^{q^*} F^*(x) dx. \quad (38)$$

Subtracting gives

$$\Delta_{ES}(G) := \text{ES}_\alpha(G) - \text{ES}_\alpha(P^*). \quad (39)$$

$$\Delta_{ES}(G) = \Delta \quad (40)$$

$$- \frac{1}{\alpha} \left[\int_{-\infty}^{q^*} g(x) dx + \int_{q^*}^{q^* + \Delta} G(x) dx \right]. \quad (41)$$

The second integral satisfies

$$\int_{q^*}^{q^* + \Delta} G(x) dx = \alpha\Delta + O(\Delta^2) + O(\eta|\Delta|), \quad (42)$$

where the last term bounds the perturbation contribution on I_δ . Since $|\Delta| \leq \eta/c$, the Δ terms cancel and the remainder is $O(\eta^2)$:

$$\text{ES}_\alpha(G) - \text{ES}_\alpha(P^*) = -\frac{1}{\alpha} \int_{-\infty}^{q^*} g(x) dx + O(\eta^2). \quad (43)$$

D. Proof of Proposition 2

We prove the VaR statement and the ES statement is identical. By Assumption 2,

$$U_t^q = b_q + \xi_t^q, \quad \mathbb{E}[\xi_t^q] = 0. \quad (44)$$

Thus

$$\frac{1}{K_S} \sum_{t \in S} U_t^q = b_q + \frac{1}{K_S} \sum_{t \in S} \xi_t^q. \quad (45)$$

Squaring and taking expectations gives

$$\mathbb{E} \left[\left(\frac{1}{K_S} \sum_{t \in S} U_t^q \right)^2 \right] = b_q^2 + \mathbb{E} \left[\left(\frac{1}{K_S} \sum_{t \in S} \xi_t^q \right)^2 \right], \quad (46)$$

because the cross term vanishes. Writing $S = \{t_0 + rM : r = 0, \dots, K_S - 1\}$, weak stationarity gives

$$\text{Cov}(\xi_{t_0+rM}^q, \xi_{t_0+sM}^q) = \sigma_q^2 \rho_q(|r-s|M). \quad (47)$$

Therefore,

$$\Xi_q := \frac{1}{K_S} \sum_{r=0}^{K_S-1} \xi_{t_0+rM}^q. \quad (48)$$

$$\begin{aligned} \mathbb{E}[\Xi_q^2] &= \frac{\sigma_q^2}{K_S^2} \sum_{r=0}^{K_S-1} \sum_{s=0}^{K_S-1} \rho_q(|r-s|M) \\ &= \frac{\sigma_q^2}{K_S} \left[1 + 2 \sum_{\ell=1}^{K_S-1} \left(1 - \frac{\ell}{K_S} \right) \rho_q(\ell M) \right] \\ &= \frac{\sigma_q^2}{K_{\text{eff},q}(K_S, M)}. \end{aligned} \quad (49)$$

This proves the VaR identity. Replacing U_t^q by U_t^{ES} proves the ES identity.

E. Proof of Proposition 3

Condition on the selected checkpoint distributions. For any fixed x ,

$$\hat{F}_{S,N}(x) = \frac{1}{K_S} \sum_{t \in S} \hat{F}_{t,m}(x), \quad (50)$$

$$\hat{F}_{t,m}(x) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{X_{t,i} \leq x\}. \quad (51)$$

Because $\mathbb{E}[\hat{F}_{t,m}(x) \mid P_t] = F_t(x)$,

$$\mathbb{E}[\hat{F}_{S,N}(x) \mid \{P_t : t \in S\}] = \frac{1}{K_S} \sum_{t \in S} F_t(x) = \bar{F}_S(x). \quad (52)$$

Conditional independence across samples and checkpoints gives

$$\begin{aligned} \text{Var}[\hat{F}_{S,N}(x) \mid \{P_t : t \in S\}] &= \frac{1}{K_S^2} \sum_{t \in S} \text{Var}[\hat{F}_{t,m}(x) \mid P_t] \\ &= \frac{1}{K_S^2 m} \sum_{t \in S} F_t(x)(1 - F_t(x)). \end{aligned} \quad (53)$$

If $m = N/K_S$, this becomes

$$\frac{1}{K_S N} \sum_{t \in S} F_t(x)(1 - F_t(x)). \quad (54)$$

Since $p \mapsto p(1 - p)$ is concave on $[0, 1]$, Jensen's inequality yields

$$\frac{1}{K_S} \sum_{t \in S} F_t(x)(1 - F_t(x)) \leq \bar{F}_S(x)(1 - \bar{F}_S(x)). \quad (55)$$

For the integrated identity, fix q . For any sample value X ,

$$\int_{-\infty}^q \mathbf{1}\{X \leq x\} dx = (q - X)_+. \quad (56)$$

Therefore

$$Z_{ES,N}(q) = \frac{1}{K_S} \sum_{t \in S} \left[\frac{1}{m} \sum_{i=1}^m (q - X_{t,i})_+ - \mathbb{E}_{P_t}(q - X_t)_+ \right]. \quad (57)$$

The conditional mean is zero, and conditional independence gives

$$\text{Var}[Z_{ES,N}(q) \mid \{P_t : t \in S\}] = \frac{1}{K_S^2 m} \sum_{t \in S} \text{Var}_{P_t}[(q - X_t)_+]. \quad (58)$$

F. Proof of Main Paper Theorem 1 (Theorem 1)

We prove VaR first. Decompose

$$\hat{q}_{\alpha,S,N}^{DSI} - q^* = D_{2,q} + D_{1,q}, \quad (59)$$

where $D_{2,q} := q_S - q^*$ and $D_{1,q} := \hat{q}_{\alpha,S,N}^{DSI} - q_S$. For the trajectory term, Lemma 1 applied to $G = \bar{F}_S$ gives

$$\begin{aligned} D_{2,q} &= -\frac{\bar{F}_S(q^*) - F^*(q^*)}{f^*(q^*)} + r_{q,S}^{traj} \\ &= -\frac{1}{f^*(q^*)} \frac{1}{K_S} \sum_{t \in S} U_t^q + r_{q,S}^{traj}, \end{aligned} \quad (60)$$

where the deterministic remainder is negligible in the sense of Assumption 4. Proposition 2 then yields

$$\mathbb{E}[D_{2,q}^2] = \frac{1}{f^*(q^*)^2} \left(b_q^2 + \frac{\sigma_q^2}{K_{\text{eff},q}(K_S, M)} \right) + o(L_q^{traj}). \quad (61)$$

For the sampling term, Assumption 5 gives

$$D_{1,q} = -\frac{Z_{q,N}}{\bar{f}_S(q_S)} + r_{q,S,N}^{sam}, \quad (62)$$

$$Z_{q,N} := \hat{F}_{S,N}(q_S) - \alpha. \quad (63)$$

Conditional on the trajectory, $\mathbb{E}[Z_{q,N} \mid \{P_t\}] = 0$ because $\bar{F}_S(q_S) = \alpha$, and Proposition 3 gives

$$\text{Var}[Z_{q,N} \mid \{P_t\}] = A_{q,N}^{\text{mix}}(S). \quad (64)$$

Therefore

$$\mathbb{E}[D_{1,q}^2] = \mathbb{E} \left[\frac{A_{q,N}^{\text{mix}}(S)}{\bar{f}_S(q_S)^2} \right] + o(N_S^{-1}) = L_{q,N}^{sam} + o(N_S^{-1}). \quad (65)$$

The leading sampling term has conditional mean zero, so

$$\mathbb{E}[D_{1,q} D_{2,q}] = \mathbb{E}[D_{2,q} \mathbb{E}(r_{q,S,N}^{sam} \mid \{P_t\})]. \quad (66)$$

By Cauchy-Schwarz, Assumption 4, and Appendix Assumption 5, this cross term is negligible relative to $L_q^{traj} + L_{q,N}^{sam}$. Combining the three pieces proves the VaR expansion.

For ES, use the analogous decomposition

$$\widehat{\text{ES}}_{\alpha,S,N}^{DSI} - \text{ES}_\alpha(P^*) = D_{2,ES} + D_{1,ES}. \quad (67)$$

where $D_{2,ES} := \text{ES}_\alpha(\bar{P}_S) - \text{ES}_\alpha(P^*)$ and $D_{1,ES} := \widehat{\text{ES}}_{\alpha,S,N}^{DSI} - \text{ES}_\alpha(\bar{P}_S)$. Lemma 1 gives

$$\begin{aligned} D_{2,ES} &= -\frac{1}{\alpha} \int_{-\infty}^{q^*} (\bar{F}_S(x) - F^*(x)) dx + r_{ES,S}^{traj} \\ &= -\frac{1}{\alpha} \frac{1}{K_S} \sum_{t \in S} U_t^{ES} + r_{ES,S}^{traj}, \end{aligned} \quad (68)$$

where the deterministic remainder is negligible in the sense of Assumption 4. Proposition 2 yields

$$\mathbb{E}[D_{2,ES}^2] = \frac{1}{\alpha^2} \left(b_{ES}^2 + \frac{\sigma_{ES}^2}{K_{\text{eff},ES}(K_S, M)} \right) + o(L_{ES}^{traj}). \quad (69)$$

For the sampling term, Assumption 5 gives

$$D_{1,ES} = -\frac{1}{\alpha} Z_{ES,N}(q_S) + r_{ES,S,N}^{sam}, \quad (70)$$

$$Z_{ES,N}(q_S) := \int_{-\infty}^{q_S} (\hat{F}_{S,N}(x) - \bar{F}_S(x)) dx. \quad (71)$$

By Proposition 3, conditional on the trajectory,

$$\mathbb{E}[Z_{ES,N}(q_S) \mid \{P_t\}] = 0, \quad (72)$$

$$\text{Var}[Z_{ES,N}(q_S) \mid \{P_t\}] = A_{ES,N}^{\text{mix}}(S). \quad (73)$$

Thus

$$\mathbb{E}[D_{1,ES}^2] = \frac{1}{\alpha^2} \mathbb{E} A_{ES,N}^{\text{mix}}(S) + o(N_S^{-1}). \quad (74)$$

The cross term is again negligible because the leading sampling term has conditional mean zero, the population remainder is controlled by Assumption 4, and the sampling remainder is mean-square negligible by Appendix Assumption 5. Combining trajectory, sampling, and cross terms proves the ES expansion.

G. Proof of Main Paper Corollary 1 (Corollary 1)

The regimes follow directly from Theorem 1. If b^2 is small and K_{eff} grows, the checkpoint-fluctuation term σ^2/K_{eff} decreases, producing a variance-reducible regime. If $K_{\text{eff}} \ll K_S$ because autocorrelation is persistent, the variance term does not decrease much, producing a correlation-limited regime. If b^2 dominates the other terms, increasing K_{eff} or N cannot remove the dominant error, producing a bias-limited regime.

H. Proof of Main Paper Proposition 1 (Proposition 1)

If $\rho(\ell M) = 1$ for every $\ell = 1, \dots, K_S - 1$, then

$$\begin{aligned}\kappa_\rho(K_S, M) &= 1 + 2 \sum_{\ell=1}^{K_S-1} \left(1 - \frac{\ell}{K_S}\right) \\ &= 1 + 2 \left[(K_S - 1) - \frac{1}{K_S} \frac{(K_S - 1)K_S}{2} \right] = K_S.\end{aligned}\quad (75)$$

Thus $K_{\text{eff}} = K_S/K_S = 1$.

For shared directional tail misspecification, suppose that for some $s \in \{-1, 1\}$,

$$s\{F_t(x) - F^*(x)\} \geq a(x) \quad \text{for every } t \in S. \quad (76)$$

Then by linearity of the mixture CDF,

$$\begin{aligned}s\{\bar{F}_S(x) - F^*(x)\} &= \frac{1}{K_S} \sum_{t \in S} s\{F_t(x) - F^*(x)\} \\ &\geq \frac{1}{K_S} \sum_{t \in S} a(x) = a(x).\end{aligned}\quad (77)$$

Thus the mixture inherits any common directional lower-tail error, including both undercoverage and overcoverage.

Finally, the bias floors follow by taking $K_{\text{eff}} \rightarrow \infty$ and $N \rightarrow \infty$ in Theorem 1. The checkpoint-fluctuation and finite-sampling terms vanish, leaving $b_q^2/f^*(q^*)^2$ for VaR and b_{ES}^2/α^2 for ES, up to negligible first-order remainders.

This appendix section complements the experimental-results section of the main paper by providing an additional dependence diagnostic for the post-burn-in checkpoint tail-error process. The main manuscript reports the regime tables and finite-budget decomposition directly there; here we retain the autocorrelation-decay visualization used to interpret effective checkpoint counts.

The autocorrelation-decay visualization is now reported directly in the main text. This appendix section therefore serves only as a pointer to the dependence diagnostic used to interpret effective checkpoint counts.

In this section, we provide the formal procedural steps for the DSI framework and detail the experimental configurations.

Algorithmic details of DSI inference and burn-in selection are summarized in the main paper's implementation-details subsection. This appendix section focuses on supplementary setup details for strategies, synthetic generators, and diffusion hyperparameters.

I. Benchmark Trading Strategies

We evaluate tail risk using a set of J benchmark trading strategies adapted from [1]. Each strategy acts as a deterministic functional mapping a multivariate asset price trajectory $\mathbf{S}_{1:T} = \{\mathbf{S}_t\}_{t=1}^T \in \mathbb{R}^{d \times T}$ (derived from the generated return sequence) to a univariate portfolio profit-and-loss (PnL) sequence.

Crucially, the terminal PnL over the investment horizon physically instantiates the random variable X introduced in the main paper's problem-formulation section, upon which

the statistical risk functionals $\text{VaR}_\alpha(X)$ and $\text{ES}_\alpha(X)$ are evaluated. To ensure strict causal evaluation, strategies are applied identically to empirical reference data and synthetic trajectories, utilizing only information available up to time t .

1) *Buy-and-Hold (Multivariate Dependence)*: The Buy-and-Hold strategy represents a static allocation baseline. Because it aggregates simultaneous price movements without dynamic trading, it explicitly tests the generative model's ability to preserve heavy-tailed multivariate dependence. Given a fixed portfolio weight vector $\mathbf{w} \in \mathbb{R}^d$ (derived from a predefined allocation matrix), the portfolio value V_t at time t is computed as the linear combination of the individual asset prices $S_{i,t}$:

$$V_t = \mathbf{w}^\top \mathbf{S}_t = \sum_{i=1}^d w_i S_{i,t}. \quad (78)$$

The strategy takes an initial position and holds it over the entire investment horizon. The terminal PnL is simply the difference between the final and initial portfolio values: $X = V_T - V_1$.

2) *Mean Reversion (Short-Term Temporal Dependence)*: The Mean Reversion (MR) strategy stress-tests the model's preservation of short-term negative autocorrelation, operating on the assumption that asset prices will eventually revert to a historical anchor. For a given asset, we define the anchoring mean μ_{init} as the average price over an initial lookback window of length W :

$$\mu_{\text{init}} = \frac{1}{W} \sum_{k=1}^W S_k. \quad (79)$$

At each subsequent active time step $t > W$, a normalized deviation score z_t is computed to measure the divergence of the current price from this initial mean:

$$z_t = \frac{S_t - \mu_{\text{init}}}{c}, \quad (80)$$

where $c > 0$ is a fixed empirical scaling factor (e.g., $c = 0.01$). Trading signals are generated dynamically based on empirical quantile thresholds. The strategy initiates a **Long** position when $z_t < \tau_{\text{lower}}$ (under the assumption that the asset is undervalued and will revert upward) and initiates a **Short** position when $z_t > \tau_{\text{upper}}$ (identifying overvaluation).

3) *Trend Following (Long-Term Temporal Dependence)*: The Trend Following (TF) strategy implements a Moving Average Crossover system to capitalize on market momentum, thereby probing the long-term positive autocorrelation of the generated paths. For a given asset, we compute two simple moving averages (SMA): a short-term average over a window of length W , and a long-term average over a window of length $2W$:

$$\text{SMA}_{\text{short},t} = \frac{1}{W} \sum_{k=0}^{W-1} S_{t-k}, \quad \text{SMA}_{\text{long},t} = \frac{1}{2W} \sum_{k=0}^{2W-1} S_{t-k}. \quad (81)$$

The trading signal is derived from the normalized divergence between these two temporal smoothing filters:

$$z_t = \frac{\text{SMA}_{\text{short},t} - \text{SMA}_{\text{long},t}}{c}. \quad (82)$$

Similar to the MR strategy, active market positions are determined by comparing z_t against upper (τ_{upper}) and lower (τ_{lower}) thresholds. However, the trading logic is inverted to follow momentum: a **Long** position is initiated when the short-term average crosses significantly above the long-term average ($z_t > \tau_{\text{upper}}$), while a **Short** position is initiated when it crosses significantly below ($z_t < \tau_{\text{lower}}$).

J. Setup of Parameters in the Synthetic Dataset

We follow the synthetic data-generation procedure of [1] exactly to ensure a fair and controlled comparison across methods.

For each time step $t \in \{1, \dots, T\}$, we first sample a latent innovation vector

$$\mathbf{u}_t = (u_{1,t}, \dots, u_{5,t})^\top \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad (83)$$

where $\Sigma \in \mathbb{R}^{5 \times 5}$ is a covariance matrix. In addition, we sample independent chi-square random variables $v_{1,t} \sim \chi^2(\nu_1)$ and $v_{2,t} \sim \chi^2(\nu_2)$, which are independent of $\mathbf{u}_t$.

The five asset price increments are then generated according to

$$\begin{aligned} \Delta p_{1,t} &= u_{1,t}, \\ \Delta p_{2,t} &= \phi_1 \Delta p_{2,t-1} + u_{2,t}, \\ \Delta p_{3,t} &= \phi_2 \Delta p_{3,t-1} + u_{3,t}, \\ \Delta p_{4,t} &= \varepsilon_{4,t} = \sigma_{4,t} \eta_{1,t}, \\ \Delta p_{5,t} &= \varepsilon_{5,t} = \sigma_{5,t} \eta_{2,t}, \end{aligned}$$

where the conditional variances evolve according to GARCH(1,1) dynamics:

$$\begin{aligned} \sigma_{4,t}^2 &= \gamma_4 + \kappa_4 \varepsilon_{4,t-1}^2 + \beta_4 \sigma_{4,t-1}^2, \\ \sigma_{5,t}^2 &= \gamma_5 + \kappa_5 \varepsilon_{5,t-1}^2 + \beta_5 \sigma_{5,t-1}^2. \end{aligned}$$

The standardized innovations are defined as

$$\eta_{1,t} = \frac{u_{4,t}}{\sqrt{v_{1,t}/\nu_1}}, \quad \eta_{2,t} = \frac{u_{5,t}}{\sqrt{v_{2,t}/\nu_2}}, \quad (84)$$

which induce heavy-tailed marginal behavior.

We set $T = 100$, corresponding to the number of intraday observations within one trading day. A correlation matrix R is first generated by sampling each element independently from $\text{Unif}(0, 1)$. Annualized standard deviations s_i are sampled uniformly from $[0.3, 0.5]$, and the covariance matrix Σ is constructed as

$$\Sigma_{ij} = \frac{s_i}{\sqrt{255T}} \frac{s_j}{\sqrt{255T}} R_{ij}, \quad i, j = 1, \dots, 5. \quad (85)$$

The remaining parameters are fixed as follows: $\phi_1 = 0.5$, $\phi_2 = -0.15$, $\nu_1 = 5$, $\nu_2 = 10$, $\kappa_4, \kappa_5 \sim \text{Unif}(0.08, 0.12)$, $\beta_4, \beta_5 \sim \text{Unif}(0.825, 0.875)$, and $\gamma_4, \gamma_5 \sim \text{Unif}(0.03, 0.07)$. Main results are reported for the risk level $\alpha = 0.05$; additional sensitivity results for $\alpha \in \{0.01, 0.05, 0.10\}$ are reported in the main paper.

K. Diffusion Parameters

Following the Denoising Diffusion Probabilistic Model (DDPM) framework [2]. The construction consists of a forward diffusion process that gradually adds noise to the data and a reverse process parameterized by a neural network to denoise it.

a) *Diffusion Process.*: We define a forward process $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ which adds Gaussian noise to the data $\mathbf{x}_0$ over $T_{\text{diff}} = 1000$ timesteps according to a variance schedule β_t :

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}). \quad (86)$$

We utilize a linear noise schedule where β_t increases linearly from $\beta_1 = 10^{-4}$ to $\beta_{T_{\text{diff}}} = 0.02$.

b) *Network Architecture.*: The reverse process is modeled by a 1D U-Net architecture, denoted as ϵ_θ , which takes the noisy input $\mathbf{x}_t \in \mathbb{R}^{d \times L}$ (where d is the number of assets and L is the time horizon) and the timestep t to predict the added noise. The network consists of:

- 1) **Time Embedding**: A sinusoidal position embedding followed by a Multi-Layer Perceptron (MLP) with SiLU activation, projecting the discrete timestep t into a 256-dimensional embedding vector.
- 2) **Encoder**: A series of 1D residual blocks and downsampling layers. Each residual block comprises two 1D convolutional layers with Group Normalization (groups=8), SiLU activations, and a time embedding projection that is added to the intermediate feature map. Downsampling is performed via strided 1D convolutions.
- 3) **Bottleneck**: A sequence of residual blocks processing the latent representation at the lowest resolution.
- 4) **Decoder**: A symmetric path of upsampling layers (transposed 1D convolutions) and residual blocks. Skip connections concatenate feature maps from the encoder to the decoder to preserve high-frequency details.

The model was trained to minimize the mean squared error (MSE) between the true noise ϵ and the predicted noise $\epsilon_\theta(\mathbf{x}_t, t)$:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2], \quad (87)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$. The network uses a base channel dimension of 64 and is optimized using AdamW with a learning rate of 10^{-5} .

This section provides supplementary empirical results that support and contextualize the main findings in the main paper. Tail-level sensitivity results are reported in the main text.

Stride	Number of Checkpoints						
	5	10	20	50	100	150	200
1	4.80 ± 0.46	4.74 ± 0.34	4.39 ± 0.17	3.84 ± 0.11	3.90 ± 0.15	4.00 ± 0.16	3.83 ± 0.17
2	4.91 ± 0.63	4.23 ± 0.41	3.73 ± 0.21	3.85 ± 0.22	3.80 ± 0.15	3.77 ± 0.12	3.94 ± 0.13
5	4.76 ± 0.62	4.47 ± 0.38	3.92 ± 0.25	3.84 ± 0.12	3.81 ± 0.19	3.85 ± 0.10	3.71 ± 0.14
10	4.71 ± 0.79	4.19 ± 0.25	4.06 ± 0.18	4.10 ± 0.27	3.66 ± 0.12	3.69 ± 0.19	3.82 ± 0.12
20	4.68 ± 0.49	4.24 ± 0.52	4.09 ± 0.31	3.95 ± 0.25	3.94 ± 0.16	4.05 ± 0.13	3.81 ± 0.12

TABLE I

FULL SENSITIVITY ANALYSIS RESULTS: THE REPORTED METRIC IS THE AVERAGE RE OF VaR AND ES IN % AT ALPHA = 5%.

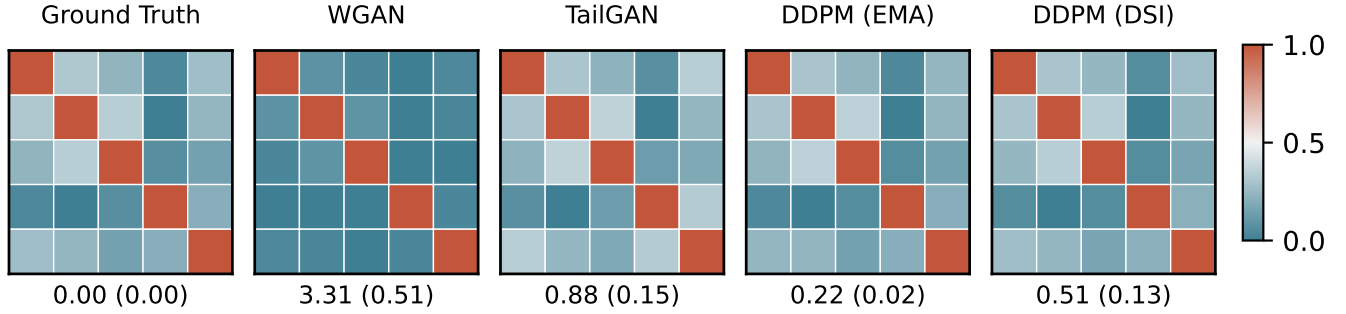


Fig. 1. Multivariate correlations of the price increments in the Synthetic data and from different trained models. Numbers below each heatmap: mean and standard deviation (in parentheses) of the sum of the absolute difference between the correlation coefficients computed with all training samples and 1,000 generated samples.

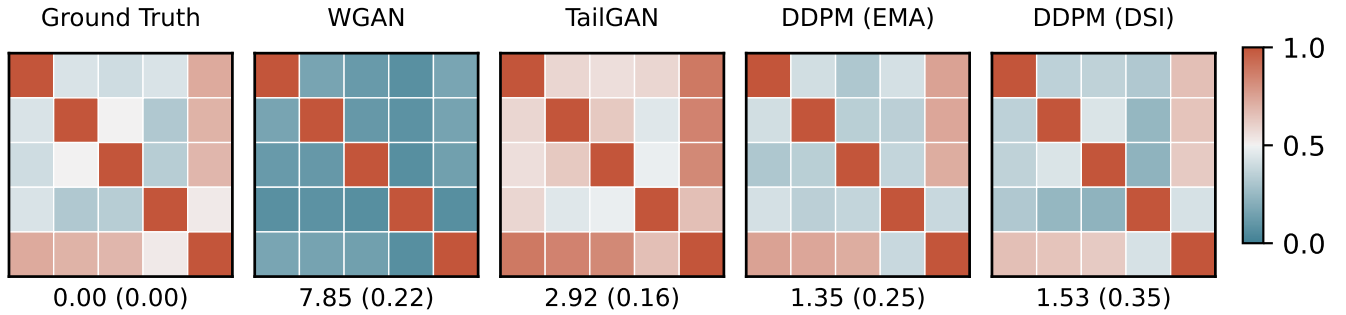


Fig. 2. Cross-asset correlations of the price increments in the Market data and from different trained models. Numbers below each heatmap: mean and standard deviation (in parentheses) of the sum of the absolute difference between the correlation coefficients computed with all training samples and 1,000 generated samples.

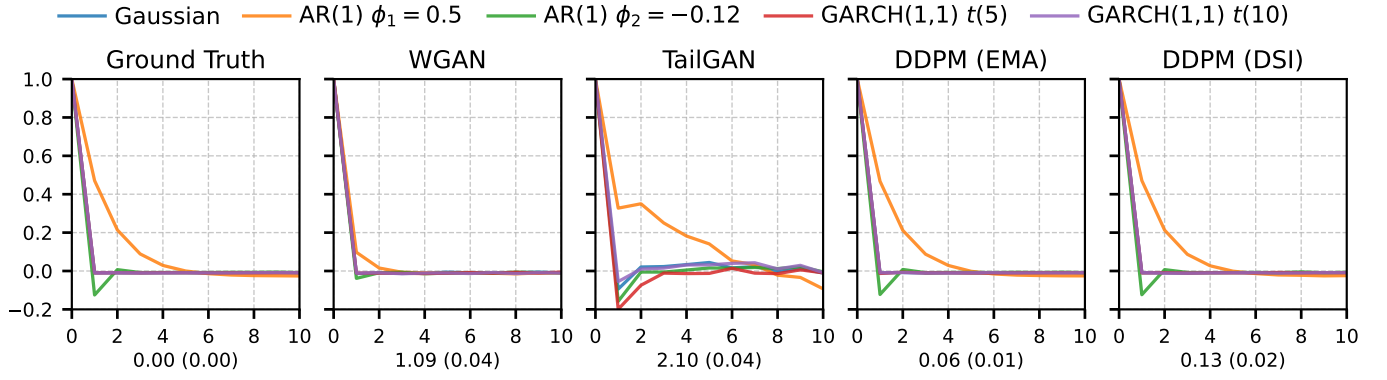


Fig. 3. Auto-correlations of the price increments from different trained models for Synthetic data. Numbers below each graph: mean and standard deviation (in parentheses) of the sum of the absolute element-wise difference between auto-correlation coefficients computed with all training samples and 1,000 generated samples.

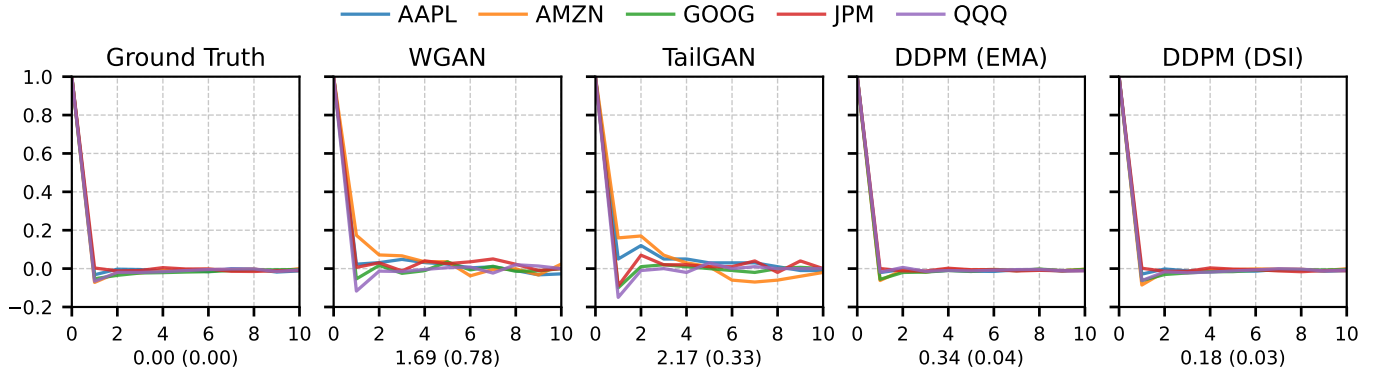


Fig. 4. Auto-correlations of the price increments from different trained models for Market data. Numbers below each graph: mean and standard deviation (in parentheses) of the sum of the absolute element-wise difference between auto-correlation coefficients computed with all training samples and 1,000 generated samples.

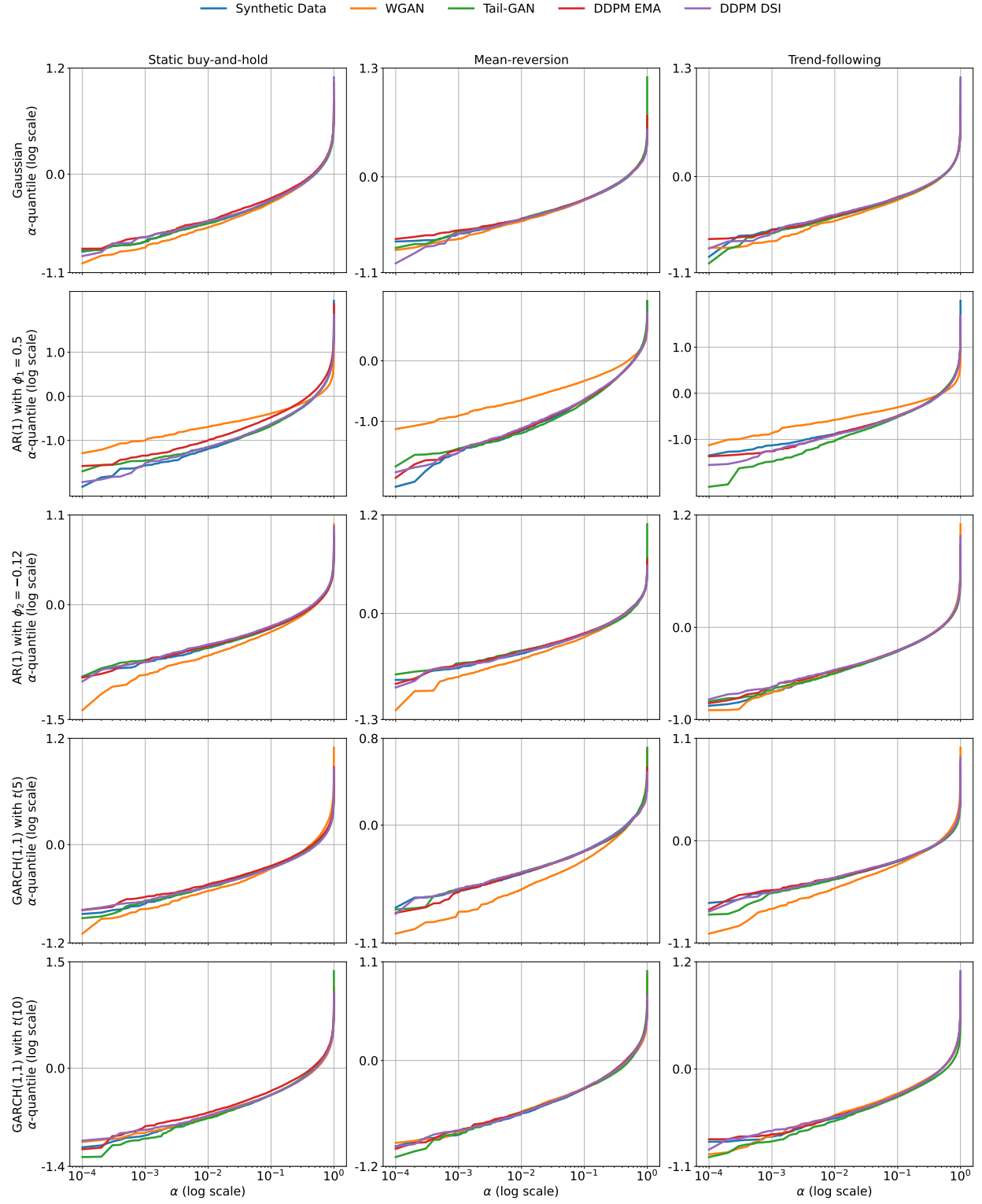


Fig. 5. Tail behavior via the synthetic rank-frequency distribution of the strategy PnL. The columns represent the strategy types.

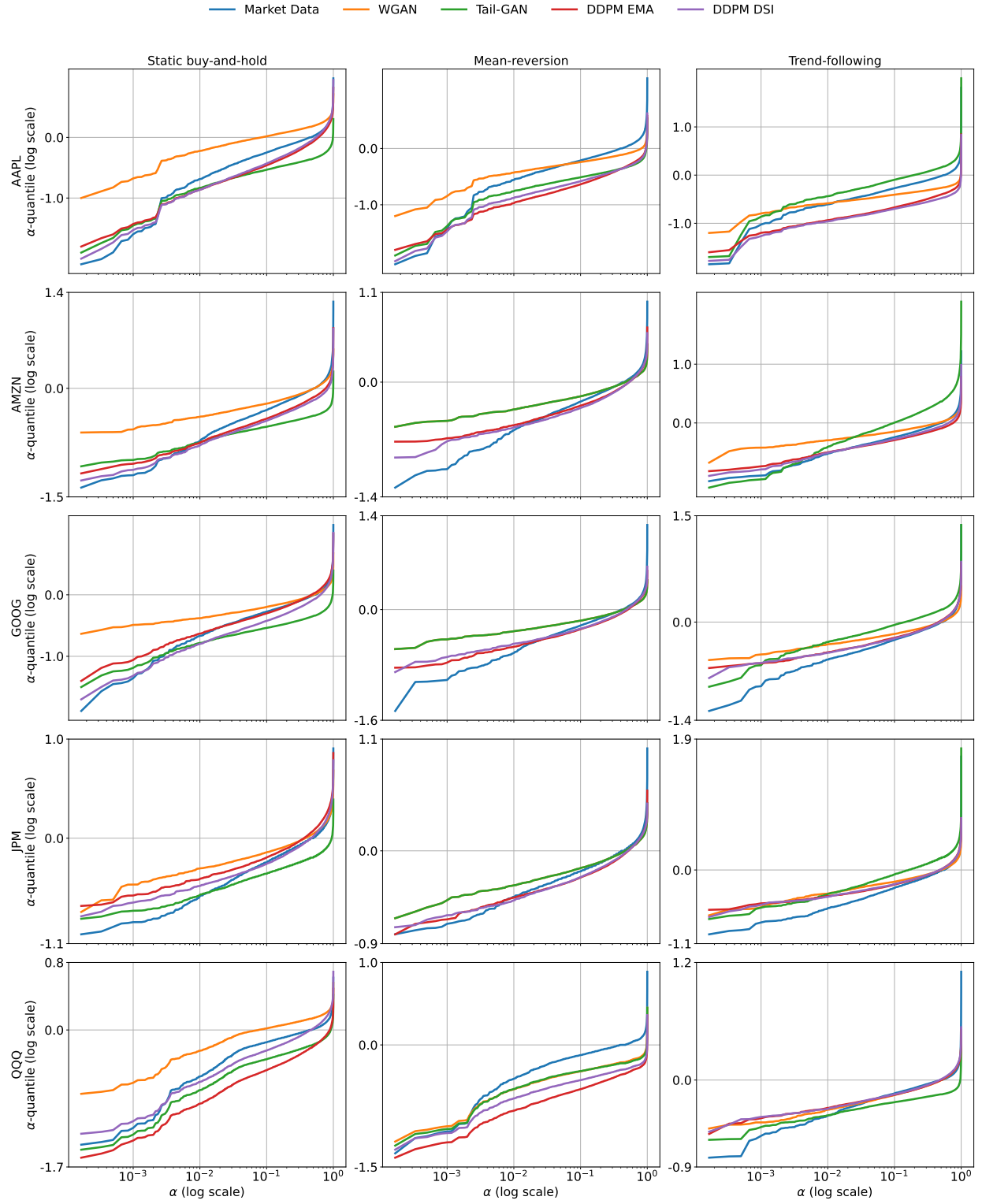


Fig. 6. Tail behavior via the market rank-frequency distribution of the strategy PnL. The columns represent the strategy types.

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- [2] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.